

Reflexion

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Reflexion: Language Agents with Verbal Reinforcement Learning Authors: Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, Shunyu Yao
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Abstract: Large language models (LLMs) have been increasingly used to interact with external environments as goal-driven agents. However, it remains challenging for these language agents to quickly and efficiently learn from trial-and-error as traditional reinforcement learning methods require extensive training samples and expensive model fine-tuning. We propose Reflexion, a novel framework to reinforce language agents not by updating weights, but instead through linguistic feedback.

Core Method: Reflexion is a framework that enables language agents to learn from failures through verbal self-reflection, without updating model weights. The key innovation is replacing scalar reward signals with natural language feedback.

The Reflexion architecture consists of three components: 1. Actor: Generates task solutions based on observations and long-term memory 2. Evaluator: Scores the actor's outputs and provides feedback signals 3. Self-Reflection: Generates verbal reinforcement cues based on task feedback

The learning process: 1. The Actor attempts a task and produces a trajectory 2. The Evaluator assesses the outcome (success/failure, scalar reward) 3. The Self-Reflection module generates natural language reflection on what went wrong 4. The reflection is stored in an episodic memory buffer 5. In subsequent trials, the Actor uses accumulated reflections to improve decision-making

Key Features: - No gradient updates required - learning happens purely through context - Supports various feedback types: scalar values, free-form language, external or internal signals - Maintains persistent memory across multiple trials - Can be combined with any base agent architecture (ReAct, chain-of-thought, etc.)

Experiments: Reflexion was evaluated on three domains: 1. Decision-Making (ALFWorld): Achieved 91% success rate, significantly outperforming ReAct baseline (73%) 2. Reasoning (HotpotQA): Improved accuracy by 13% over baseline through iterative self-correction 3. Programming (HumanEval, LeetCode): Achieved state-of-the-art pass@1 rates, with 80% on HumanEval and 91% on LeetCode Hard

Key Findings: - Verbal reinforcement is more efficient than scalar rewards for language agents - Self-reflection enables rapid learning from just a few failures - The approach is model-agnostic and can be applied to any LLM - Reflexion demonstrates emergent generalization across related tasks